

CanSat Ground Control Station (GCS)



INDIA SPACE LAB

Project Report

17th June to 31st July 2026

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CANSAT GROUND CONTROL STATION

Real-time Telemetry Dashboard Documentation

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INDIA SPACE LAB (ISL) METEOROLOGY & FLIGHT CONTROL INTERNSHIP

Purpose

This single-page Ground Control Station (GCS) supports a CanSat mission by ingesting telemetry, monitoring flight health, displaying location and attitude, issuing mission commands, and exporting mission data. The implementation follows the CanSat assignment PDF; the module order is used only to organize development.

Requirements coverage

PDF requirement	Implementation
Mission dashboard and control bar	Responsive HTML/CSS dashboard with start/stop, reset, sync, calibration, health, and export controls.
Mission controls	Confirmation-protected separation, emergency parachute, and redundant activation commands.
Continuous telemetry	Physics-based simulator and optional Web Serial receiver accepting JSON or CSV packets.
Fault monitoring	Live four-digit descent-rate, GPS, separation, and parachute indicator.
Five real-time plots	Chart.js altitude, temperature, pressure, descent-rate, and battery charts.
GPS tracking	Leaflet/OpenStreetMap marker, coordinates, and bounded 300-point trajectory.

Orientation	Canvas artificial horizon using roll, pitch, and yaw.
Video	Browser camera selection and start/stop feed.
Data management	Packet table, 100-entry console, CSV export, and combined graph PNG export.

How to Run & Test

1. **Launcher:** Double-click `Launch\_GCS.bat` to open the Ground Control Station dashboard in your default web browser.
2. **Internet Connection:** Ensure your machine has an active internet connection to load external Leaflet maps and Chart.js dependencies.
3. **Start Telemetry:** Keep the telemetry connection source selector on **SIMULATOR - Physics Engine** and click **START TELEMTRY** to initialize the flight.
4. **Observe Flight Phases:** Observe the metrics panel updates. You will see the altitude climb during ascent, separation trigger at apogee (700m), descent speed adjustments, and landing confirmation.
5. **Verify Subsystems:**
 - Check the ****GPS Map**** to see the red flight path updates around Haldwani.
 - Check the ****Real-Time Telemetry Plots**** updating instantly on the second row.
 - Check the ****Orientation Gauge**** rendering roll, pitch, and yaw canvas tilt.
 - Select your camera and click ****START VIDEO**** to allow browser camera access and verify the live webcam card.
6. **Trigger Critical Commands:** Click **MANUAL SEPARATION** and **DEPLOY EMERGENCY PARACHUTE** buttons. Confirm the browser prompt dialogs and monitor warning logs.
7. **Diagnostic Error Bytes:** Watch the 4-digit binary indicators (`digit-1` to `digit-4`) update live (e.g., descent rate flags, GPS flags, and parachute indicators).
8. **Export Data:** Click **EXPORT LOGS** to download the CSV telemetry database, and click **EXPORT GRAPHS** to save a graphic snapshot of the charts.

Serial telemetry format

The GCS accepts a newline-delimited JSON telemetry object. A CSV packet is also accepted in this order:

`packet,altitude,descentRate,temperature,pressure,battery,latitude,longitude,roll,pitch,yaw`

Example JSON packet:

```
```json
```

```
{"packet":1,"altitude":125.5,"descentRate":0,"temperature":26.3,"pressure":99828,"battery":8.39,"latitude":28.613918,"longitude":77.209022,"roll":4.8,"pitch":-2.3,"yaw":12,"gps":true}
```

```
```
```

Serial command format

For a connected CanSat, command messages are newline-delimited JSON. Firmware must parse this format and respond using the telemetry protocol.

```
```json
```

```
{"type":"command","command":"MANUAL_SEPARATION","timestamp":"2026-07-23T00:00:00.000Z","missionTime":32,"packet":8}
```

```
```
```

Supported commands are `MANUAL\_SEPARATION`, `EMERGENCY\_PARACHUTE`, and `REDUNDANT\_ACTIVATION`.

Verification performed

- JavaScript syntax checks on every module.
- DOM ID and chart-canvas reference checks.
- CSS brace-balance check.
- Sample telemetry JSON validation.

Hardware verification remains necessary before flight use: validate the exact serial protocol, baud rate, sensors, command acknowledgements, GPS lock, camera permission, and recovery mechanisms with the supplied microcontroller kit.

Demonstration Video

A professional video walkthrough demonstrating the functional features of this Ground Control Station is included in the package:

- **Location**: `Demonstration/CanSat\_GCS\_Demonstration.mp4`
- **Content Covered**:
 - Launching the dashboard using `Launch\_GCS.bat`.
 - Activating the telemetry simulation stream and watching real-time plots.

- Verifying coordinates path tracking on the GPS Map.
- Verifying attitude pitch and roll on the Artificial Horizon canvas.
- Initiating critical command separation and emergency parachute prompts.
- Syncing clock times and resetting telemetry packet logs.